

Question 12.14

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1 Question:

4-point DFT of a real discrete-time signal $x[n]$ of length 4 is given by $X[k]$, $n = 0, 1, 2, 3$ and $k = 0, 1, 2, 3$. It is given that $X[0] = 5$, $X[1] = 1 + j1$, $X[2] = 0.5$. $X[3]$ and $x[0]$ respectively are

a) $1 - j$, 1.875

c) $1 + j$, 1.875

b) $1 - j$, 1.500

d) $0.1 - j0.1$, 1.500

2 Solution:

Since our sequence is of length $n = 4$, let us denote $\omega_n = e^{-j\frac{2\pi}{4}} = -j$. The DFT matrix thus formed is:

$$W_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega_n & \omega_n^2 & \omega_n^3 \\ 1 & \omega_n^2 & \omega_n^4 & \omega_n^6 \\ 1 & \omega_n^3 & \omega_n^6 & \omega_n^9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix}$$

The inverse DFT matrix is given by:

$$W_4^{-1} = \frac{1}{4}W_4^H = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{pmatrix}$$

(Here A^H denotes the Hermitian transpose of matrix A, ie the conjugate transpose of A). To get the DFT of our sequence, we use the formula: $\mathbf{X} = W_4\mathbf{x}$ or $\mathbf{x} = W_4^{-1}\mathbf{X}$ to get our sequence from our DFT.

We know $X = \begin{pmatrix} 5 \\ 1+j \\ 0.5 \\ X[3] \end{pmatrix}$. Calculating $x[0]$:

$$x[0] = \frac{1}{4}(5 + (1+j) + 0.5 + X[3]) = \frac{6.5 + j + X[3]}{4}$$

Since $x[0]$ is real, the imaginary part must be zero. Thus, the imaginary part of $X[3]$ must be $-j$. Therefore, $X[3] = a - j$ where a is a real number. Calculating $x[1]$:

$$x[1] = \frac{1}{4}(5 + j(1+j) - 0.5 - j(a-j)) = \frac{4.5 + (1-a)j}{4}$$

Calculating $x[2]$:

$$x[2] = \frac{1}{4}(5 - (1+j) + 0.5 + (a-j)) = \frac{4.5 + (a-1) - 2j}{4}$$

Calculating $x[3]$:

$$x[3] = \frac{1}{4}(5 - j(1+j) - 0.5 + j(a-j)) = \frac{4.5 + (a-1)j}{4}$$

Since $x[n]$ is real for all values of n , the imaginary parts of $x[1]$ and $x[3]$ must be zero. Thus, we have:

$$1 - a = 0 \implies a = 1$$

Thus, $X[3] = 1 - j$. Calculating $x[0]$:

$$x[0] = \frac{6.5 + j + (1-j)}{4} = \frac{7.5}{4} = 1.875$$

Thus, the correct option is (a) $X[3] = 1 - j$ and $x[0] = 1.875$.